Ref.	Cont.	Variants	Mitigations	Proof	Security	Platform
[14]	$[\![\cdot]\!]^{\text{spec}}_{\text{ct}}$	PHT, RSB	FI, SLH	(✓)	●	S
[3]	$[\![\cdot]\!]^{\text{spec}}_{\text{ct}}$	PHT, RSB	CMR	×	◒	HS
[17]	$[\![\cdot]\!]^{\text{spec}}_{\text{arch}}$	PHT	DTT	(✓)	◒	H
[4]	?	PHT	FI	×	●	S
[4]	?	SSB	MSR	×	●	S/H
[7]	$[\![\cdot]\!]^{\text{spec}}_{\text{ct}}$	PHT, BTB, RSB, STL, PSF	FI,ISO,RZ	(✓)	●	S
[8]	$[\![\cdot]\!]^{\text{spec}}_{\text{mem}}$	PHT, BTB, RSB, STL, PSF	ISO,SM,RI	×	●	S/H
[1]	$[\![\cdot]\!]^{\text{spec}}_{\text{mem}}$	PHT, BTB, SSB, STL	-	×	◒	HS
[15]	?	PHT, BTB, RSB, SSB, STL	-	×	●	HS
[11]	?	PHT, BTB, RSB, SSB, STL	NTMMap	×	●	S/H
[5]	$[\![\cdot]\!]^{\text{spec}}_{\text{ct}}$	PHT, BTB	SM	×	●	H
[12]	$[\![\cdot]\!]^{\text{spec}}_{\text{ct}}$	BTB, RSB	FI,ISO	×	●	S
[6]	$[\![\cdot]\!]^{\text{spec}}_{\text{ct}}$	BTB, RSB	CFI	×	●	S/H
[16]	$[\![\cdot]\!]^{\text{spec}}_{\text{ct}}$	PHT, BTB, RSB, SSB, STL	ISO	×	●	H
[9]	?	PHT	ADD	×	?	S
[2]	?	PHT	SLH	×	◒	S
[10]	$[\![\cdot]\!]^{\text{spec}}_{\text{ct}}$	PHT	SSLH	✓	◒	S
[18]	$[\![\cdot]\!]^{\text{spec}}_{\text{ct}}$	PHT	USLH	✓	●	S
[13]	$[\![\cdot]\!]^{\text{spec}}_{\text{ct}}$	PHT, RSB, BTB	FI	×	●	H

[0]The above table uses execution modes Sequential, in-order execution of instructions. (SEQ), Sequential, out-of-order execution of instructions. (SEQOOO), Speculative, in-order execution of instructions. (SPEC), and Speculative, out-of-order execution of instructions. (SPECOOO). Moreover, it relies on observers Sees loads, stores, and program counter memory addresses. (CTobs), Constant-Time observer. Sees loads, stores, and program counter memory addresses, as well as all possible (speculative) paths. (PATH-CTobs), Sees loads and store memory addresses. (MEMobs), and Architectural observer. Sees any microarchitectural covert channel. (ARCHobs).

1 Preliminaries

For all traces, we assume that if $\lightning$ occurs, it occurs exactly once and no events follow after. The $\lightning$ event mimicks a crash.

▷ **LEMMA** 1 (Left-total and injective $\sim$ induces a Galois Insertion.). If

(a) $\sim$ left total

(b) $\sim$ left unique

then

(i) $\forall \pi, \widetilde{\varsigma}\left(\widetilde{\tau}\left(\pi\right)\right) = \pi$

Proof.

Let π be a set of source-level objects. Let $\mathsf{a_S}$ be a source-level object. The proof goes by antisymmetry of set-inclusion:

Case $\subseteq$:

We know:

(H_1) $\mathsf{a_S} \in \widetilde{\varsigma}\left(\widetilde{\tau}\left(\pi\right)\right)$

We need to show:

(i) $\mathsf{a_S} \in \pi$

From Assumption (a) we know that there is a $\mathbf{a_T}$ such that:

(H_1) $\mathsf{a_S} \sim \mathbf{a_T}$

Specialize Assumption (H_1) with Assumption (H_1), obtaining:

(H_1) $\mathbf{a_T} \in \widetilde{\tau}\left(\pi\right)$

From Assumption (H_1), we know that there is a $\mathsf{a'_S}$ such that:

(H_1) $\mathsf{a'_S} \sim \mathbf{a_T}$

(H_2) $\mathsf{a'_S} \in \pi$

Now, Goal (i) follows from Assumption (H_2) by rewriting with Assumption (b) whose assumptions are satisfied via Assumption (H_1) and Assumption (H_1).

Case $\supseteq$:

We know:

(H_3) $\mathsf{a_S} \in \pi$

We need to show:

(i) $\mathsf{a_S} \in \widetilde{\varsigma}\left(\widetilde{\tau}\left(\pi\right)\right)$

Unfold Goal (i), so let $\mathbf{a_T}$ such that:

(H_1) $\mathsf{a_S} \sim \mathbf{a_T}$

What is left to prove is:

(i) $\mathbf{a_T} \in \widetilde{\tau}(\pi)$

Simply instantiate the existential in Goal (i) with $a^{\text{seq}}_{\text{arch}}$. Assumption (H_1) and Assumption (H_3) resolve the remaining obligations.

□

2 Observers

$$
\begin{array}{lcl}
[\![\cdot]\!]^{\text{seq}}_{\text{ct}} & - & a^{\text{seq}}_{\text{ct}} = \ulcorner \varepsilon \urcorner^{\text{seq}}_{\text{ct}} \mid \ulcorner \lightning \urcorner^{\text{seq}}_{\text{ct}} \mid \ulcorner \mathit{Load}\ \ell \urcorner^{\text{seq}}_{\text{ct}} \mid \ulcorner \mathit{Store}\ \ell \urcorner^{\text{seq}}_{\text{ct}} \mid \ulcorner \mathit{Pc}\ n \urcorner^{\text{seq}}_{\text{ct}} \\
[\![\cdot]\!]^{\text{spec}}_{\text{ct}} & - & a^{\text{spec}}_{\text{ct}} = \ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{ct}} \mid \ulcorner \lightning \urcorner^{\text{spec}}_{\text{ct}} \mid \ulcorner \mathit{Load}\ \ell \urcorner^{\text{spec}}_{\text{ct}} \mid \ulcorner \mathit{Store}\ \ell \urcorner^{\text{spec}}_{\text{ct}} \mid \ulcorner \mathit{Pc}\ n \urcorner^{\text{spec}}_{\text{ct}} \mid \ulcorner \mathit{Spec} \urcorner^{\text{spec}}_{\text{ct}} \mid \ulcorner \mathit{Rlb} \urcorner^{\text{spec}}_{\text{ct}} \\
[\![\cdot]\!]^{\text{seq}}_{\text{mem}} & - & a^{\text{seq}}_{\text{mem}} = \ulcorner \varepsilon \urcorner^{\text{seq}}_{\text{mem}} \mid \ulcorner \lightning \urcorner^{\text{seq}}_{\text{mem}} \mid \ulcorner \mathit{Load}\ \ell \urcorner^{\text{seq}}_{\text{mem}} \mid \ulcorner \mathit{Store}\ \ell \urcorner^{\text{seq}}_{\text{mem}} \\
[\![\cdot]\!]^{\text{spec}}_{\text{mem}} & - & a^{\text{spec}}_{\text{mem}} = \ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{mem}} \mid \ulcorner \lightning \urcorner^{\text{spec}}_{\text{mem}} \mid \ulcorner \mathit{Load}\ \ell \urcorner^{\text{spec}}_{\text{mem}} \mid \ulcorner \mathit{Store}\ \ell \urcorner^{\text{spec}}_{\text{mem}} \mid \ulcorner \mathit{Spec} \urcorner^{\text{spec}}_{\text{mem}} \mid \ulcorner \mathit{Rlb} \urcorner^{\text{spec}}_{\text{mem}} \\
[\![\cdot]\!]^{\text{seq}}_{\text{arch}} & - & a^{\text{seq}}_{\text{arch}} = \ulcorner \varepsilon \urcorner^{\text{seq}}_{\text{arch}} \mid \ulcorner \lightning \urcorner^{\text{seq}}_{\text{arch}} \mid \ulcorner \mathit{Load}\ \ell\ v \urcorner^{\text{seq}}_{\text{arch}} \mid \ulcorner \mathit{Store}\ \ell\ v \urcorner^{\text{seq}}_{\text{arch}} \mid \ulcorner \mathit{Pc}\ n \urcorner^{\text{seq}}_{\text{arch}} \\
[\![\cdot]\!]^{\text{spec}}_{\text{arch}} & - & a^{\text{spec}}_{\text{arch}} = \ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{arch}} \mid \ulcorner \lightning \urcorner^{\text{spec}}_{\text{arch}} \mid \ulcorner \mathit{Load}\ \ell\ v \urcorner^{\text{spec}}_{\text{arch}} \mid \ulcorner \mathit{Store}\ \ell\ v \urcorner^{\text{spec}}_{\text{arch}} \mid \ulcorner \mathit{Pc}\ n \urcorner^{\text{spec}}_{\text{arch}} \mid \ulcorner \mathit{Spec} \urcorner^{\text{spec}}_{\text{arch}} \mid \ulcorner \mathit{Rlb} \urcorner^{\text{spec}}_{\text{arch}}
\end{array}
$$

3 Mappings between Observers

3.1 Sequential to Speculative

All relations presented in this section share the same key idea, so it is mostly repetitive. The key insight is to index the relations by natural numbers in order to keep track of the depth of nested speculation and ignore if the depth is non-zero, i.e., if there is speculation. The trace-level versions have to do minor bookkeeping to count speculation depth, but the counting is done at event-level.

Constant-Time

$\boxed{\delta^{\text{seqct}}_{\text{specct}} : \ulcorner \ell \urcorner^{\text{seq}}_{\text{ct}} \to \ulcorner \ell \urcorner^{\text{spec}}_{\text{ct}}}$ „Total function between memory locations across sequential constant-time and speculative constant-time events."

$\boxed{\sim^{\mathit{seqct}\ (n)}_{\mathit{specct}\ (m)} : \mathbb{N} \to \mathbb{N} \to a^{\text{seq}}_{\text{ct}} \to a^{\text{spec}}_{\text{ct}} \to \mathbb{P}}$ „Map sequential constant time to speculative constant time events. n represents the current size of speculation depth and m the new size."

$$\frac{}{\ulcorner \varepsilon \urcorner^{\text{seq}}_{\text{ct}} \sim^{\mathit{seqct}\ (n)}_{\mathit{specct}\ (n)} \ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{ct}}} \quad \text{(seqct-to-specct-empty)}$$

$$\frac{}{\ulcorner \lightning \urcorner^{\text{seq}}_{\text{ct}} \sim^{\mathit{seqct}\ (n)}_{\mathit{specct}\ (n)} \ulcorner \lightning \urcorner^{\text{spec}}_{\text{ct}}} \quad \text{(seqct-to-specct-crash)}$$

$$\frac{}{\ulcorner \mathit{Pc}\ n \urcorner^{\text{seq}}_{\text{ct}} \sim^{\mathit{seqct}\ (0)}_{\mathit{specct}\ (0)} \ulcorner \mathit{Pc}\ n \urcorner^{\text{spec}}_{\text{ct}}} \quad \text{(seqct-to-specct-pc)}$$

$$\frac{\delta^{\text{seqct}}_{\text{specct}}(\ulcorner \ell \urcorner^{\text{seq}}_{\text{ct}}) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{ct}}}{\ulcorner \mathit{Store}\ \ell \urcorner^{\text{seq}}_{\text{ct}} \sim^{\mathit{seqct}\ (0)}_{\mathit{specct}\ (0)} \ulcorner \mathit{Store}\ \ell \urcorner^{\text{spec}}_{\text{ct}}} \quad \text{(seqct-to-specct-store)}$$

$$\frac{\delta^{\text{seqct}}_{\text{specct}}(\ulcorner \ell \urcorner^{\text{seq}}_{\text{ct}}) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{ct}}}{\ulcorner \mathit{Load}\ \ell \urcorner^{\text{seq}}_{\text{ct}} \sim^{\mathit{seqct}\ (0)}_{\mathit{specct}\ (0)} \ulcorner \mathit{Load}\ \ell \urcorner^{\text{spec}}_{\text{ct}}} \quad \text{(seqct-to-specct-load)}$$

$$\frac{}{\varepsilon \sim^{\mathit{seqct}\ (n)}_{\mathit{specct}\ (1+n)} \ulcorner \mathit{Spec} \urcorner^{\text{spec}}_{\text{ct}}} \quad \text{(seqct-to-specct-spec)}$$

$$\frac{}{\varepsilon \sim^{\mathit{seqct}\ (1+n)}_{\mathit{specct}\ (n)} \ulcorner \mathit{Rlb} \urcorner^{\text{spec}}_{\text{ct}}} \quad \text{(seqct-to-specct-rlb)}$$

$$\frac{}{\ulcorner \varepsilon \urcorner^{\text{seq}}_{\text{ct}} \sim^{\mathit{seqct}\ (1+n)}_{\mathit{specct}\ (1+n)} \ulcorner \mathit{Pc}\ m \urcorner^{\text{spec}}_{\text{ct}}} \quad \text{(seqct-to-specct-pc-ign)}$$

$$\frac{\delta^{\text{seqct}}_{\text{specct}}(\ulcorner \ell \urcorner^{\text{seq}}_{\text{ct}}) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{ct}}}{\ulcorner \varepsilon \urcorner^{\text{seq}}_{\text{ct}} \sim^{\mathit{seqct}\ (1+n)}_{\mathit{specct}\ (1+n)} \ulcorner \mathit{Store}\ \ell \urcorner^{\text{spec}}_{\text{ct}}} \quad \text{(seqct-to-specct-store-ign)}$$

$$\frac{\delta^{\text{seqct}}_{\text{specct}}(\ulcorner \ell \urcorner^{\text{seq}}_{\text{ct}}) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{ct}}}{\ulcorner \varepsilon \urcorner^{\text{seq}}_{\text{ct}} \sim^{\mathit{seqct}\ (1+n)}_{\mathit{specct}\ (1+n)} \ulcorner \mathit{Load}\ \ell \urcorner^{\text{spec}}_{\text{ct}}} \quad \text{(seqct-to-specct-load-ign)}$$

$\approx^{seqct\ (n)}_{specct\ (m)} : \mathbb{N} \to \mathbb{N} \to \overline{a^{\text{seq}}_{\text{ct}}} \to \overline{a^{\text{spec}}_{\text{ct}}} \to \mathbb{P}$ „Map sequential constant time to speculative constant time events. n represents the current size of speculation depth and m the new size."

$$\frac{}{[\cdot] \approx^{seqct\ (0)}_{specct\ (0)} [\cdot]}\ \textsf{(seqct-to-specct-refl)} \qquad \frac{a^{\text{seq}}_{\text{ct}} \sim^{seqct\ (n)}_{specct\ (m')} a^{\text{spec}}_{\text{ct}} \qquad \overline{a^{\text{seq}}_{\text{ct}}} \approx^{seqct\ (m')}_{specct\ (m)} \overline{a^{\text{spec}}_{\text{ct}}}}{a^{\text{seq}}_{\text{ct}} \cdot \overline{a^{\text{seq}}_{\text{ct}}} \approx^{seqct\ (n)}_{specct\ (m)} a^{\text{spec}}_{\text{ct}} \cdot \overline{a^{\text{spec}}_{\text{ct}}}}\ \textsf{(seqct-to-specct-trans)}$$

$$\frac{a^{\text{seq}}_{\text{ct}} \sim^{seqct\ (n)}_{specct\ (m')} \ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{ct}} \qquad \overline{a^{\text{seq}}_{\text{ct}}} \approx^{seqct\ (m')}_{specct\ (m)} \overline{a^{\text{spec}}_{\text{ct}}}}{a^{\text{seq}}_{\text{ct}} \cdot \overline{a^{\text{seq}}_{\text{ct}}} \approx^{seqct\ (n)}_{specct\ (m)} \overline{a^{\text{spec}}_{\text{ct}}}}\ \textsf{(seqct-to-specct-ignR)}$$

$$\frac{\ulcorner \varepsilon \urcorner^{\text{seq}}_{\text{ct}} \sim^{seqct\ (n)}_{specct\ (m')} a^{\text{spec}}_{\text{ct}} \qquad \overline{a^{\text{seq}}_{\text{ct}}} \approx^{seqct\ (m')}_{specct\ (m)} \overline{a^{\text{spec}}_{\text{ct}}}}{\overline{a^{\text{seq}}_{\text{ct}}} \approx^{seqct\ (n)}_{specct\ (m)} a^{\text{spec}}_{\text{ct}} \cdot \overline{a^{\text{spec}}_{\text{ct}}}}\ \textsf{(seqct-to-specct-ignL)}$$

▷ **LEMMA** 2 ($\sim^{seqct\ (n)}_{specct\ (m)}$ IS LEFT-UNIQUE.). $\sim^{seqct\ (n)}_{specct\ (m)}$ is left-unique.

PROOF. Let $a^{\text{seq}}_{\text{ct}}$ and $a^{\text{seq}'}_{\text{ct}}$ with $a^{\text{spec}}_{\text{ct}}$ such that:

(H_1) $a^{\text{seq}}_{\text{ct}} \sim^{seqct\ (n)}_{specct\ (m)} a^{\text{spec}}_{\text{ct}}$

(H_2) $a^{\text{seq}'}_{\text{ct}} \sim^{seqct\ (n)}_{specct\ (m)} a^{\text{spec}}_{\text{ct}}$

The goal is:

(i) $a^{\text{seq}}_{\text{ct}} = a^{\text{seq}'}_{\text{ct}}$

Invert both Assumption (H_1) and Assumption (H_2), then Goal (i) follows by reflexivity. □

▷ **LEMMA** 3 ($\sim^{seqct\ (n)}_{specct\ (m)}$ IS LEFT-TOTAL.). $\sim^{seqct\ (n)}_{specct\ (m)}$ is left-total.

PROOF. Simple case analysis on the arguments. □

▷ **COROLLARY** 1 ($\sim^{seqct\ (n)}_{specct\ (m)}$ INDUCES A GALOIS INSERTION.). It holds that:

(i) $\forall \pi, \tilde{\varsigma}\left(\tilde{\tau}(\pi)\right) = \pi$

PROOF. Immediate from Lemma 1 (Left-total and injective $\sim$ induces a Galois Insertion.) via Lemma 3 ($\sim^{seqct\ (n)}_{specct\ (m)}$ is left-total.) and Lemma 2 ($\sim^{seqct\ (n)}_{specct\ (m)}$ is left-unique.). □

Memory

$\delta^{\text{seqmem}}_{\text{specmem}} : \ulcorner \ell \urcorner^{\text{seq}}_{\text{mem}} \to \ulcorner \ell \urcorner^{\text{spec}}_{\text{mem}}$ „Total function between memory locations across sequential memory and speculative memory events."

$\sim^{seqmem\ (n)}_{specmem\ (m)} : \mathbb{N} \to \mathbb{N} \to a^{\text{seq}}_{\text{mem}} \to a^{\text{spec}}_{\text{mem}} \to \mathbb{P}$ „Map sequential memory to speculative memory events. n represents the current size of speculation depth and m the new size."

$$\frac{}{\ulcorner \varepsilon \urcorner^{\text{seq}}_{\text{mem}} \sim^{seqmem\ (n)}_{specmem\ (n)} \ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{mem}}}\ \textsf{(seqmem-to-specmem-empty)} \qquad \frac{}{\ulcorner \lightning \urcorner^{\text{seq}}_{\text{mem}} \sim^{seqmem\ (n)}_{specmem\ (n)} \ulcorner \lightning \urcorner^{\text{spec}}_{\text{mem}}}\ \textsf{(seqmem-to-specmem-crash)}$$

$$\frac{\delta^{\text{seqmem}}_{\text{specmem}}\left(\ulcorner \ell \urcorner^{\text{seq}}_{\text{mem}}\right) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{mem}}}{\ulcorner Store\ \ell \urcorner^{\text{seq}}_{\text{mem}} \sim^{seqmem\ (0)}_{specmem\ (0)} \ulcorner Store\ \ell \urcorner^{\text{spec}}_{\text{mem}}}\ \textsf{(seqmem-to-specmem-store)}$$

$$\frac{\delta^{\text{seqmem}}_{\text{specmem}}\left(\ulcorner \ell \urcorner^{\text{seq}}_{\text{mem}}\right) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{mem}}}{\ulcorner Load\ \ell \urcorner^{\text{seq}}_{\text{mem}} \sim^{seqmem\ (0)}_{specmem\ (0)} \ulcorner Load\ \ell \urcorner^{\text{spec}}_{\text{mem}}}\ \textsf{(seqmem-to-specmem-load)}$$

$$\frac{}{\varepsilon \sim^{seqmem\ (n)}_{specmem\ (1+n)} \ulcorner Spec \urcorner^{\text{spec}}_{\text{mem}}}\ \textsf{(seqmem-to-specmem-spec)} \qquad \frac{}{\varepsilon \sim^{seqmem\ (1+n)}_{specmem\ (n)} \ulcorner Rlb \urcorner^{\text{spec}}_{\text{mem}}}\ \textsf{(seqmem-to-specmem-rlb)}$$

$$\frac{\delta^{\text{seqmem}}_{\text{specmem}}\left(\ulcorner \ell \urcorner^{\text{seq}}_{\text{mem}}\right) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{mem}}}{\ulcorner \varepsilon \urcorner^{\text{seq}}_{\text{mem}} \sim^{\mathit{seqmem}\,(1+n)}_{\mathit{specmem}\,(1+n)} \ulcorner \mathit{Store}\ \ell \urcorner^{\text{spec}}_{\text{mem}}} \quad \text{(seqmem-to-specmem-store-ign)}$$

$$\frac{\delta^{\text{seqmem}}_{\text{specmem}}\left(\ulcorner \ell \urcorner^{\text{seq}}_{\text{mem}}\right) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{mem}}}{\ulcorner \varepsilon \urcorner^{\text{seq}}_{\text{mem}} \sim^{\mathit{seqmem}\,(1+n)}_{\mathit{specmem}\,(1+n)} \ulcorner \mathit{Load}\ \ell \urcorner^{\text{spec}}_{\text{mem}}} \quad \text{(seqmem-to-specmem-load-ign)}$$

$\approx^{\mathit{seqmem}\,(n)}_{\mathit{specmem}\,(m)} : \mathbb{N} \to \mathbb{N} \to \overline{a^{\text{seq}}_{\text{mem}}} \to \overline{a^{\text{spec}}_{\text{mem}}} \to \mathbb{P}$ „Map sequential memory to speculative memory events. n represents the current size of speculation depth and m the new size."

$$\frac{}{[\cdot] \approx^{\mathit{seqmem}\,(0)}_{\mathit{specmem}\,(0)} [\cdot]} \quad \text{(seqmem-to-specmem-refl)}$$

$$\frac{a^{\text{seq}}_{\text{mem}} \sim^{\mathit{seqmem}\,(n)}_{\mathit{specmem}\,(m')} a^{\text{spec}}_{\text{mem}} \qquad \overline{a^{\text{seq}}_{\text{mem}}} \approx^{\mathit{seqmem}\,(m')}_{\mathit{specmem}\,(m)} \overline{a^{\text{spec}}_{\text{mem}}}}{a^{\text{seq}}_{\text{mem}} \cdot \overline{a^{\text{seq}}_{\text{mem}}} \approx^{\mathit{seqmem}\,(n)}_{\mathit{specmem}\,(m)} a^{\text{spec}}_{\text{mem}} \cdot \overline{a^{\text{spec}}_{\text{mem}}}} \quad \text{(seqmem-to-specmem-trans)}$$

$$\frac{a^{\text{seq}}_{\text{mem}} \sim^{\mathit{seqmem}\,(n)}_{\mathit{specmem}\,(m')} \ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{mem}} \qquad \overline{a^{\text{seq}}_{\text{mem}}} \approx^{\mathit{seqmem}\,(m')}_{\mathit{specmem}\,(m)} \overline{a^{\text{spec}}_{\text{mem}}}}{a^{\text{seq}}_{\text{mem}} \cdot \overline{a^{\text{seq}}_{\text{mem}}} \approx^{\mathit{seqmem}\,(n)}_{\mathit{specmem}\,(m)} \overline{a^{\text{spec}}_{\text{mem}}}} \quad \text{(seqmem-to-specmem-ignR)}$$

$$\frac{\ulcorner \varepsilon \urcorner^{\text{seq}}_{\text{mem}} \sim^{\mathit{seqmem}\,(n)}_{\mathit{specmem}\,(m')} a^{\text{spec}}_{\text{mem}} \qquad \overline{a^{\text{seq}}_{\text{mem}}} \approx^{\mathit{seqmem}\,(m')}_{\mathit{specmem}\,(m)} \overline{a^{\text{spec}}_{\text{mem}}}}{\overline{a^{\text{seq}}_{\text{mem}}} \approx^{\mathit{seqmem}\,(n)}_{\mathit{specmem}\,(m)} a^{\text{spec}}_{\text{mem}} \cdot \overline{a^{\text{spec}}_{\text{mem}}}} \quad \text{(seqmem-to-specmem-ignL)}$$

▷ **LEMMA** 4 ($\sim^{\mathit{seqmem}\,(n)}_{\mathit{specmem}\,(m)}$ IS LEFT-UNIQUE.). $\sim^{\mathit{seqmem}\,(n)}_{\mathit{specmem}\,(m)}$ is left-unique.

PROOF. Let $a^{\text{seq}}_{\text{mem}}$ and $a^{\text{seq}\,\prime}_{\text{mem}}$ with $a^{\text{spec}}_{\text{mem}}$ such that:

(H_1) $a^{\text{seq}}_{\text{mem}} \sim^{\mathit{seqmem}\,(n)}_{\mathit{specmem}\,(m)} a^{\text{spec}}_{\text{mem}}$

(H_2) $a^{\text{seq}\,\prime}_{\text{mem}} \sim^{\mathit{seqmem}\,(n)}_{\mathit{specmem}\,(m)} a^{\text{spec}}_{\text{mem}}$

The goal is:

(i) $a^{\text{seq}}_{\text{mem}} = a^{\text{seq}\,\prime}_{\text{mem}}$

Invert both Assumption (H_1) and Assumption (H_2), then Goal (i) follows by reflexivity. □

▷ **LEMMA** 5 ($\sim^{\mathit{seqmem}\,(n)}_{\mathit{specmem}\,(m)}$ IS LEFT-TOTAL.). $\sim^{\mathit{seqmem}\,(n)}_{\mathit{specmem}\,(m)}$ is left-total.

PROOF. Simple case analysis on the arguments. □

▷ **COROLLARY** 2 ($\sim^{\mathit{seqmem}\,(n)}_{\mathit{specmem}\,(m)}$ INDUCES A GALOIS INSERTION.). It holds that:

(i) $\forall \pi, \tilde{\varsigma}\left(\tilde{\tau}\left(\pi\right)\right) = \pi$

PROOF. Immediate from Lemma 1 (Left-total and injective $\sim$ induces a Galois Insertion.) via Lemma 5 ($\sim^{\mathit{seqmem}\,(n)}_{\mathit{specmem}\,(m)}$ is left-total.) and Lemma 4 ($\sim^{\mathit{seqmem}\,(n)}_{\mathit{specmem}\,(m)}$ is left-unique.). □

Architecture

$\delta^{\text{seqarch}}_{\text{specarch}} : \ulcorner \ell \urcorner^{\text{seq}}_{\text{arch}} \to \ulcorner \ell \urcorner^{\text{spec}}_{\text{arch}}$ „Total function between archory locations across sequential architecture and speculative architecture events."

$\sim^{\mathit{seqarch}\,(n)}_{\mathit{specarch}\,(m)} : \mathbb{N} \to \mathbb{N} \to a^{\text{seq}}_{\text{arch}} \to a^{\text{spec}}_{\text{arch}} \to \mathbb{P}$ „Map sequential architecture to speculative architecture events. n represents the current size of speculation depth and m the new size."

$$\frac{}{\ulcorner \varepsilon \urcorner^{\text{seq}}_{\text{arch}} \sim^{\mathit{seqarch}\,(n)}_{\mathit{specarch}\,(n)} \ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{arch}}} \quad \text{(seqarch-to-specarch-empty)} \qquad \frac{}{\ulcorner \lightning \urcorner^{\text{seq}}_{\text{arch}} \sim^{\mathit{seqarch}\,(n)}_{\mathit{specarch}\,(n)} \ulcorner \lightning \urcorner^{\text{spec}}_{\text{arch}}} \quad \text{(seqarch-to-specarch-crash)}$$

$$\frac{}{\ulcorner Pc\ n \urcorner^{\text{seq}}_{\text{arch}} \sim^{seqarch\ (0)}_{specarch\ (0)} \ulcorner Pc\ n \urcorner^{\text{spec}}_{\text{arch}}} \quad \text{(seqarch-to-specarch-pc)}$$

$$\frac{\delta^{\text{seqarch}}_{\text{specarch}}\left(\ulcorner \ell \urcorner^{\text{seq}}_{\text{arch}}\right) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{arch}}}{\ulcorner Store\ \ell\ v \urcorner^{\text{seq}}_{\text{arch}} \sim^{seqarch\ (0)}_{specarch\ (0)} \ulcorner Store\ \ell\ v \urcorner^{\text{spec}}_{\text{arch}}} \quad \text{(seqarch-to-specarch-store)}$$

$$\frac{\delta^{\text{seqarch}}_{\text{specarch}}\left(\ulcorner \ell \urcorner^{\text{seq}}_{\text{arch}}\right) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{arch}}}{\ulcorner Load\ \ell\ v \urcorner^{\text{seq}}_{\text{arch}} \sim^{seqarch\ (0)}_{specarch\ (0)} \ulcorner Load\ \ell\ v \urcorner^{\text{spec}}_{\text{arch}}} \quad \text{(seqarch-to-specarch-load)}$$

$$\frac{}{\varepsilon \sim^{seqarch\ (n)}_{specarch\ (1+n)} \ulcorner Spec \urcorner^{\text{spec}}_{\text{arch}}} \quad \text{(seqarch-to-specarch-spec)} \qquad \frac{}{\varepsilon \sim^{seqarch\ (1+n)}_{specarch\ (n)} \ulcorner Rlb \urcorner^{\text{spec}}_{\text{arch}}} \quad \text{(seqarch-to-specarch-rlb)}$$

$$\frac{}{\ulcorner \varepsilon \urcorner^{\text{seq}}_{\text{arch}} \sim^{seqarch\ (1+n)}_{specarch\ (1+n)} \ulcorner Pc\ m \urcorner^{\text{spec}}_{\text{arch}}} \quad \text{(seqarch-to-specarch-pc-ign)}$$

$$\frac{\delta^{\text{seqarch}}_{\text{specarch}}\left(\ulcorner \ell \urcorner^{\text{seq}}_{\text{arch}}\right) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{arch}}}{\ulcorner \varepsilon \urcorner^{\text{seq}}_{\text{arch}} \sim^{seqarch\ (1+n)}_{specarch\ (1+n)} \ulcorner Store\ \ell\ v \urcorner^{\text{spec}}_{\text{arch}}} \quad \text{(seqarch-to-specarch-store-ign)}$$

$$\frac{\delta^{\text{seqarch}}_{\text{specarch}}\left(\ulcorner \ell \urcorner^{\text{seq}}_{\text{arch}}\right) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{arch}}}{\ulcorner \varepsilon \urcorner^{\text{seq}}_{\text{arch}} \sim^{seqarch\ (1+n)}_{specarch\ (1+n)} \ulcorner Load\ \ell\ v \urcorner^{\text{spec}}_{\text{arch}}} \quad \text{(seqarch-to-specarch-load-ign)}$$

$$\boxed{\approx^{seqarch\ (n)}_{specarch\ (m)} : \mathbb{N} \to \mathbb{N} \to \overline{a^{\text{seq}}_{\text{arch}}} \to \overline{a^{\text{spec}}_{\text{arch}}} \to \mathbb{P}}$$

„Map sequential architecture to speculative architecture events. n represents the current size of speculation depth and m the new size."

$$\frac{}{[\cdot] \approx^{seqarch\ (0)}_{specarch\ (0)} [\cdot]} \quad \text{(seqarch-to-specarch-refl)} \qquad \frac{a^{\text{seq}}_{\text{arch}} \sim^{seqarch\ (n)}_{specarch\ (m')} a^{\text{spec}}_{\text{arch}} \qquad \overline{a^{\text{seq}}_{\text{arch}}} \approx^{seqarch\ (m')}_{specarch\ (m)} \overline{a^{\text{spec}}_{\text{arch}}}}{a^{\text{seq}}_{\text{arch}} \cdot \overline{a^{\text{seq}}_{\text{arch}}} \approx^{seqarch\ (n)}_{specarch\ (m)} a^{\text{spec}}_{\text{arch}} \cdot \overline{a^{\text{spec}}_{\text{arch}}}} \quad \text{(seqarch-to-specarch-trans)}$$

$$\frac{a^{\text{seq}}_{\text{arch}} \sim^{seqarch\ (n)}_{specarch\ (m')} \ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{arch}} \qquad \overline{a^{\text{seq}}_{\text{arch}}} \approx^{seqarch\ (m')}_{specarch\ (m)} \overline{a^{\text{spec}}_{\text{arch}}}}{a^{\text{seq}}_{\text{arch}} \cdot \overline{a^{\text{seq}}_{\text{arch}}} \approx^{seqarch\ (n)}_{specarch\ (m)} \overline{a^{\text{spec}}_{\text{arch}}}} \quad \text{(seqarch-to-specarch-ignR)}$$

$$\frac{\ulcorner \varepsilon \urcorner^{\text{seq}}_{\text{arch}} \sim^{seqarch\ (n)}_{specarch\ (m')} a^{\text{spec}}_{\text{arch}} \qquad \overline{a^{\text{seq}}_{\text{arch}}} \approx^{seqarch\ (m')}_{specarch\ (m)} \overline{a^{\text{spec}}_{\text{arch}}}}{\overline{a^{\text{seq}}_{\text{arch}}} \approx^{seqarch\ (n)}_{specarch\ (m)} a^{\text{spec}}_{\text{arch}} \cdot \overline{a^{\text{spec}}_{\text{arch}}}} \quad \text{(seqarch-to-specarch-ignL)}$$

▷ **LEMMA** 6 ($\sim^{seqarch\ (n)}_{specarch\ (m)}$ IS LEFT-UNIQUE.). $\sim^{seqarch\ (n)}_{specarch\ (m)}$ is left-unique.

PROOF. Let $a^{\text{seq}}_{\text{arch}}$ and $a^{\text{seq}\prime}_{\text{arch}}$ with $a^{\text{spec}}_{\text{arch}}$ such that:

(H_1) $a^{\text{seq}}_{\text{arch}} \sim^{seqarch\ (n)}_{specarch\ (m)} a^{\text{spec}}_{\text{arch}}$

(H_2) $a^{\text{seq}\prime}_{\text{arch}} \sim^{seqarch\ (n)}_{specarch\ (m)} a^{\text{spec}}_{\text{arch}}$

The goal is:

(i) $a^{\text{seq}}_{\text{arch}} = a^{\text{seq}\prime}_{\text{arch}}$

Invert both Assumption (H_1) and Assumption (H_2), then Goal (i) follows by reflexivity. □

▷ **LEMMA** 7 ($\sim^{seqarch\ (n)}_{specarch\ (m)}$ IS LEFT-TOTAL.). $\sim^{seqarch\ (n)}_{specarch\ (m)}$ is left-total.

PROOF. Simple case analysis on the arguments. □

▷ **COROLLARY** 3 ($\sim^{seqarch\ (n)}_{specarch\ (m)}$ INDUCES A GALOIS INSERTION.). It holds that:

(i) $\forall \pi, \tilde{\varsigma}\left(\tilde{\tau}\left(\pi\right)\right) = \pi$

PROOF. Immediate from Lemma 1 (Left-total and injective $\sim$ induces a Galois Insertion.) via Lemma 7 ($\sim^{seqarch\ (n)}_{specarch\ (m)}$

is left-total.) and Lemma 6 ($\sim^{seqarch\ (n)}_{specarch\ (m)}$ is left-unique.). □

▷ **LEMMA** 8 ($\approx^{seqarch\ (n)}_{specarch\ (m)}$ IS LEFT-UNIQUE.). $\approx^{seqarch\ (n)}_{specarch\ (m)}$ is left-unique.

PROOF. Let $\overline{a^{\text{seq}}_{\text{arch}}}$ and $\overline{a^{\text{seq}\prime}_{\text{arch}}}$ with $\overline{a^{\text{spec}}_{\text{arch}}}$ such that:

(H_1) $\overline{a^{\text{seq}}_{\text{arch}}} \approx^{seqarch\ (n)}_{specarch\ (m)} \overline{a^{\text{spec}}_{\text{arch}}}$

(H_2) $\overline{a^{\text{seq}\prime}_{\text{arch}}} \approx^{seqarch\ (n)}_{specarch\ (m)} \overline{a^{\text{spec}}_{\text{arch}}}$

The goal is:

(i) $\overline{a^{\text{seq}}_{\text{arch}}} = \overline{a^{\text{seq}\prime}_{\text{arch}}}$

Induce on Assumption (H_1) then invert Assumption (H_2). The base-case follows by reflexivity and the induction case follows by inductive hypothesis and Lemma 6 ($\sim^{seqarch\ (n)}_{specarch\ (m)}$ is left-unique.). □

▷ **LEMMA** 9 ($\approx^{seqarch\ (n)}_{specarch\ (m)}$ IS LEFT-TOTAL.). $\approx^{seqarch\ (n)}_{specarch\ (m)}$ is left-total.

PROOF. Let $\overline{a^{\text{seq}}_{\text{arch}}}$ be a trace. We want to show that there are $n, m, \overline{a^{\text{spec}}_{\text{arch}}}$ such that $\overline{a^{\text{seq}}_{\text{arch}}} \approx^{seqarch\ (n)}_{specarch\ (m)} \overline{a^{\text{spec}}_{\text{arch}}}$
Simply induce on $\overline{a^{\text{seq}}_{\text{arch}}}$ and use Lemma 7 ($\sim^{seqarch\ (n)}_{specarch\ (m)}$ is left-total.). □

▷ **COROLLARY** 4 ($\approx^{seqarch\ (n)}_{specarch\ (m)}$ INDUCES A GALOIS INSERTION.). It holds that:

(i) $\forall \pi, \underset{\varsigma}{\overset{\approx^{seqarch\ (n)}_{specarch\ (m)}}{}} \left(\underset{\tau}{\overset{\approx^{seqarch\ (n)}_{specarch\ (m)}}{}} (\pi) \right) = \pi$

PROOF. Immediate from Lemma 1 (Left-total and injective $\sim$ induces a Galois Insertion.) via Lemma 9 ($\approx^{seqarch\ (n)}_{specarch\ (m)}$ is left-total.) and Lemma 8 ($\approx^{seqarch\ (n)}_{specarch\ (m)}$ is left-unique.). □

3.2 Across Observer Types

Memory ↔ Constant-time

$\delta^{\text{specmem}}_{\text{specct}} : \ulcorner \ell \urcorner^{\text{spec}}_{\text{mem}} \to \ulcorner \ell \urcorner^{\text{spec}}_{\text{ct}}$ „Total function between locations across speculative memory and speculative constant-time events."

$\sim^{specmem}_{specct} : a^{\text{spec}}_{\text{mem}} \to a^{\text{spec}}_{\text{ct}} \to \mathbb{P}$ „Map speculative memory to speculative constant-time events.

$$\frac{}{\ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{mem}} \sim^{specmem}_{specct} \ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{ct}}} \text{(specmem-to-specct-empty)} \qquad \frac{}{\ulcorner \lightning \urcorner^{\text{spec}}_{\text{mem}} \sim^{specmem}_{specct} \ulcorner \lightning \urcorner^{\text{spec}}_{\text{ct}}} \text{(specmem-to-specct-crash)}$$

$$\frac{\delta^{\text{specmem}}_{\text{specct}} (\ulcorner \ell \urcorner^{\text{spec}}_{\text{mem}}) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{ct}}}{\ulcorner Load\ \ell \urcorner^{\text{spec}}_{\text{mem}} \sim^{specmem}_{specct} \ulcorner Load\ \ell \urcorner^{\text{spec}}_{\text{ct}}} \text{(specmem-to-specct-load)}$$

$$\frac{\delta^{\text{specmem}}_{\text{specct}} (\ulcorner \ell \urcorner^{\text{spec}}_{\text{mem}}) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{ct}}}{\ulcorner Store\ \ell \urcorner^{\text{spec}}_{\text{mem}} \sim^{specmem}_{specct} \ulcorner Store\ \ell \urcorner^{\text{spec}}_{\text{ct}}} \text{(specmem-to-specct-store)} \qquad \frac{}{\ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{mem}} \sim^{specmem}_{specct} \ulcorner Pc\ \ell \urcorner^{\text{spec}}_{\text{ct}}} \text{(specmem-to-specct-pcign)}$$

$$\frac{}{\ulcorner Spec \urcorner^{\text{spec}}_{\text{mem}} \sim^{specmem}_{specct} \ulcorner Spec \urcorner^{\text{spec}}_{\text{ct}}} \text{(specmem-to-specct-spec)} \qquad \frac{}{\ulcorner Rlb \urcorner^{\text{spec}}_{\text{mem}} \sim^{specmem}_{specct} \ulcorner Rlb \urcorner^{\text{spec}}_{\text{ct}}} \text{(specmem-to-specct-rlb)}$$

$\approx^{specmem}_{specct} : \overline{a^{\text{spec}}_{\text{mem}}} \to \overline{a^{\text{spec}}_{\text{ct}}} \to \mathbb{P}$ „Map speculative memory to speculative constant-time traces."

$$\frac{}{[\cdot] \approx^{specmem}_{specct} [\cdot]} \text{(specmem-to-specct-refl)} \qquad \frac{a^{\text{spec}}_{\text{mem}} \sim^{specmem}_{specct} a^{\text{spec}}_{\text{ct}} \qquad \overline{a^{\text{spec}}_{\text{mem}}} \approx^{specmem}_{specct} \overline{a^{\text{spec}}_{\text{ct}}}}{a^{\text{spec}}_{\text{mem}} \cdot \overline{a^{\text{spec}}_{\text{mem}}} \approx^{specmem}_{specct} a^{\text{spec}}_{\text{mem}} \cdot \overline{a^{\text{spec}}_{\text{mem}}}} \text{(specmem-to-specct-trans)}$$

$$\frac{a^{\text{spec}}_{\text{mem}} \sim^{specmem}_{specct} \ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{ct}} \qquad \overline{a^{\text{spec}}_{\text{mem}}} \approx^{specmem}_{specct} \overline{a^{\text{spec}}_{\text{ct}}}}{a^{\text{spec}}_{\text{mem}} \cdot \overline{a^{\text{spec}}_{\text{mem}}} \approx^{specmem}_{specct} \overline{a^{\text{spec}}_{\text{ct}}}} \text{(specmem-to-specct-ignR)}$$

$$\frac{\ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{mem}} \sim^{seqmem}_{specct} a^{\text{spec}}_{\text{ct}} \qquad \overline{a^{\text{spec}}_{\text{mem}}} \approx^{specmem}_{specct} \overline{a^{\text{spec}}_{\text{ct}}}}{\overline{a^{\text{spec}}_{\text{mem}}} \approx^{specmem}_{specct} a^{\text{spec}}_{\text{ct}} \cdot \overline{a^{\text{spec}}_{\text{ct}}}} \quad \text{(specmem-to-specct-ignL)}$$

Using existing relations, it is now possible to define sequential comparison without loss of precision as follows:

$$\overline{a^{\text{seq}}_{\text{mem}}} \approx^{seqmem}_{specct} \overline{a^{\text{spec}}_{\text{ct}}} \stackrel{\text{def}}{=} \exists \overline{a^{\text{spec}}_{\text{mem}}}, \overline{a^{\text{seq}}_{\text{mem}}} \approx^{seqmem\ (0)}_{specmem\ (0)} \overline{a^{\text{spec}}_{\text{mem}}} \text{ and } \overline{a^{\text{spec}}_{\text{mem}}} \approx^{specmem}_{specct} \overline{a^{\text{spec}}_{\text{ct}}}$$

$$\overline{a^{\text{seq}}_{\text{mem}}} \approx^{seqmem}_{seqct} \overline{a^{\text{seq}}_{\text{ct}}} \stackrel{\text{def}}{=} \exists \overline{a^{\text{spec}}_{\text{ct}}}, \overline{a^{\text{seq}}_{\text{mem}}} \approx^{seqmem}_{specct} \overline{a^{\text{spec}}_{\text{ct}}} \text{ and } \overline{a^{\text{seq}}_{\text{ct}}} \approx^{seqct\ (0)}_{specct\ (0)} \overline{a^{\text{spec}}_{\text{ct}}}$$

Moreover:

$$\overline{a^{\text{spec}}_{\text{mem}}} \approx^{specmem}_{seqct} \overline{a^{\text{seq}}_{\text{ct}}} \stackrel{\text{def}}{=} \exists \overline{a^{\text{spec}}_{\text{ct}}}, \overline{a^{\text{spec}}_{\text{mem}}} \approx^{specmem}_{specct} \overline{a^{\text{spec}}_{\text{ct}}} \text{ and } \overline{a^{\text{seq}}_{\text{ct}}} \approx^{seqct\ (0)}_{specct\ (0)} \overline{a^{\text{spec}}_{\text{ct}}}$$

Constant-time ↔ Architecture

$\boxed{\delta^{\text{specct}}_{\text{specarch}} : \ulcorner \ell \urcorner^{\text{spec}}_{\text{ct}} \to \ulcorner \ell \urcorner^{\text{spec}}_{\text{arch}}}$ „Total funarchion between locations across speculative constant-time and speculative architecture events."

$\boxed{\sim^{specct}_{specarch} : a^{\text{spec}}_{\text{ct}} \to a^{\text{spec}}_{\text{arch}} \to \mathbb{P}}$ „Map speculative constant-time to speculative architecture events.

$$\frac{}{\ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{ct}} \sim^{specct}_{specarch} \ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{arch}}} \quad \text{(specct-to-specarch-empty)} \qquad \frac{}{\ulcorner \lightning \urcorner^{\text{spec}}_{\text{ct}} \sim^{specct}_{specarch} \ulcorner \lightning \urcorner^{\text{spec}}_{\text{arch}}} \quad \text{(specct-to-specarch-crash)}$$

$$\frac{\delta^{\text{specct}}_{\text{specarch}} (\ulcorner \ell \urcorner^{\text{spec}}_{\text{ct}}) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{arch}}}{\ulcorner Load\ \ell \urcorner^{\text{spec}}_{\text{ct}} \sim^{specct}_{specarch} \ulcorner Load\ \ell\ v \urcorner^{\text{spec}}_{\text{arch}}} \quad \text{(specct-to-specarch-load)}$$

$$\frac{\delta^{\text{specct}}_{\text{specarch}} (\ulcorner \ell \urcorner^{\text{spec}}_{\text{ct}}) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{arch}}}{\ulcorner Store\ \ell \urcorner^{\text{spec}}_{\text{ct}} \sim^{specct}_{specarch} \ulcorner Store\ \ell\ v \urcorner^{\text{spec}}_{\text{arch}}} \quad \text{(specct-to-specarch-store)}$$

$$\frac{}{\ulcorner Pc\ n \urcorner^{\text{spec}}_{\text{ct}} \sim^{specct}_{specarch} \ulcorner Pc\ n \urcorner^{\text{spec}}_{\text{arch}}} \quad \text{(specct-to-specarch-pc)} \qquad \frac{}{\ulcorner Spec \urcorner^{\text{spec}}_{\text{ct}} \sim^{specct}_{specarch} \ulcorner Spec \urcorner^{\text{spec}}_{\text{arch}}} \quad \text{(specct-to-specarch-spec)}$$

$$\frac{}{\ulcorner Rlb \urcorner^{\text{spec}}_{\text{ct}} \sim^{specct}_{specarch} \ulcorner Rlb \urcorner^{\text{spec}}_{\text{arch}}} \quad \text{(specct-to-specarch-rlb)}$$

$\boxed{\approx^{specct}_{specarch} : \overline{a^{\text{spec}}_{\text{ct}}} \to \overline{a^{\text{spec}}_{\text{arch}}} \to \mathbb{P}}$ „Map speculative constant-time to speculative architecture traces."

$$\frac{}{[\cdot] \approx^{specct}_{specarch} [\cdot]} \quad \text{(specct-to-specarch-refl)} \qquad \frac{a^{\text{spec}}_{\text{ct}} \sim^{specct}_{specarch} a^{\text{spec}}_{\text{arch}} \qquad \overline{a^{\text{spec}}_{\text{ct}}} \approx^{specct}_{specarch} \overline{a^{\text{spec}}_{\text{arch}}}}{a^{\text{spec}}_{\text{ct}} \cdot \overline{a^{\text{spec}}_{\text{ct}}} \approx^{specct}_{specarch} a^{\text{spec}}_{\text{ct}} \cdot \overline{a^{\text{spec}}_{\text{ct}}}} \quad \text{(specct-to-specarch-trans)}$$

$$\frac{a^{\text{spec}}_{\text{ct}} \sim^{specct}_{specarch} \ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{arch}} \qquad \overline{a^{\text{spec}}_{\text{ct}}} \approx^{specct}_{specarch} \overline{a^{\text{spec}}_{\text{arch}}}}{a^{\text{spec}}_{\text{ct}} \cdot \overline{a^{\text{spec}}_{\text{ct}}} \approx^{specct}_{specarch} \overline{a^{\text{spec}}_{\text{arch}}}} \quad \text{(specct-to-specarch-ignR)}$$

$$\frac{\ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{ct}} \sim^{seqct}_{specarch} a^{\text{spec}}_{\text{arch}} \qquad \overline{a^{\text{spec}}_{\text{ct}}} \approx^{specct}_{specarch} \overline{a^{\text{spec}}_{\text{arch}}}}{\overline{a^{\text{spec}}_{\text{ct}}} \approx^{specct}_{specarch} a^{\text{spec}}_{\text{arch}} \cdot \overline{a^{\text{spec}}_{\text{arch}}}} \quad \text{(specct-to-specarch-ignL)}$$

Using existing relations, it is now possible to define sequential comparison without loss of precision as follows:

$$\overline{a^{\text{seq}}_{\text{ct}}} \approx^{seqct}_{specarch} \overline{a^{\text{spec}}_{\text{arch}}} \stackrel{\text{def}}{=} \exists \overline{a^{\text{spec}}_{\text{ct}}}, \overline{a^{\text{seq}}_{\text{ct}}} \approx^{seqct\ (0)}_{specct\ (0)} \overline{a^{\text{spec}}_{\text{ct}}} \text{ and } \overline{a^{\text{spec}}_{\text{ct}}} \approx^{specct}_{specarch} \overline{a^{\text{spec}}_{\text{arch}}}$$

$$\overline{a^{\text{seq}}_{\text{ct}}} \approx^{seqct}_{seqarch} \overline{a^{\text{seq}}_{\text{arch}}} \stackrel{\text{def}}{=} \exists \overline{a^{\text{spec}}_{\text{arch}}}, \overline{a^{\text{seq}}_{\text{ct}}} \approx^{seqct}_{specarch} \overline{a^{\text{spec}}_{\text{arch}}} \text{ and } \overline{a^{\text{seq}}_{\text{arch}}} \approx^{seqarch\ (0)}_{specarch\ (0)} \overline{a^{\text{spec}}_{\text{arch}}}$$

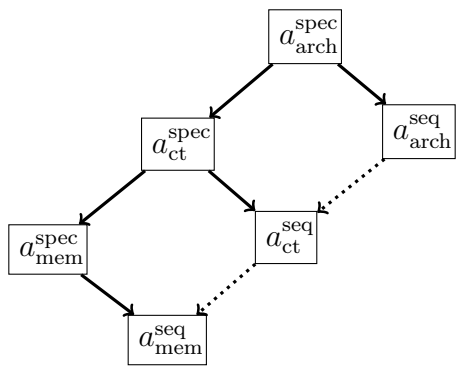
Moreover:

$$\overline{a^{\text{spec}}_{\text{ct}}} \approx^{specct}_{seqarch} \overline{a^{\text{seq}}_{\text{arch}}} \stackrel{\text{def}}{=} \exists \overline{a^{\text{spec}}_{\text{arch}}}, \overline{a^{\text{spec}}_{\text{ct}}} \approx^{specct}_{specarch} \overline{a^{\text{spec}}_{\text{arch}}} \text{ and } \overline{a^{\text{seq}}_{\text{arch}}} \approx^{seqarch\ (0)}_{specarch\ (0)} \overline{a^{\text{spec}}_{\text{arch}}}$$

Memory ↔ Architecture
These cases are plugging the previous relations into each other in a straightforward fashion:

$$\begin{array}{lcl}
\overline{a^{\text{seq}}_{\text{mem}}} \approx^{seqmem}_{seqarch} \overline{a^{\text{seq}}_{\text{arch}}} & \stackrel{\text{def}}{=} & \exists \overline{a^{\text{seq}}_{\text{ct}}}, \overline{a^{\text{seq}}_{\text{mem}}} \approx^{seqmem}_{seqct} \overline{a^{\text{seq}}_{\text{ct}}} \text{ and } \overline{a^{\text{seq}}_{\text{ct}}} \approx^{seqct}_{seqarch} \overline{a^{\text{seq}}_{\text{arch}}} \\
\overline{a^{\text{seq}}_{\text{mem}}} \approx^{seqmem}_{specarch} \overline{a^{\text{spec}}_{\text{arch}}} & \stackrel{\text{def}}{=} & \exists \overline{a^{\text{seq}}_{\text{ct}}}, \overline{a^{\text{seq}}_{\text{mem}}} \approx^{seqmem}_{seqct} \overline{a^{\text{seq}}_{\text{ct}}} \text{ and } \overline{a^{\text{seq}}_{\text{ct}}} \approx^{seqct}_{specarch} \overline{a^{\text{spec}}_{\text{arch}}} \\
\overline{a^{\text{spec}}_{\text{mem}}} \approx^{specmem}_{seqarch} \overline{a^{\text{seq}}_{\text{arch}}} & \stackrel{\text{def}}{=} & \exists \overline{a^{\text{spec}}_{\text{ct}}}, \overline{a^{\text{spec}}_{\text{mem}}} \approx^{specmem}_{specct} \overline{a^{\text{spec}}_{\text{ct}}} \text{ and } \overline{a^{\text{spec}}_{\text{ct}}} \approx^{specct}_{seqarch} \overline{a^{\text{seq}}_{\text{arch}}} \\
\overline{a^{\text{spec}}_{\text{mem}}} \approx^{specmem}_{specarch} \overline{a^{\text{spec}}_{\text{arch}}} & \stackrel{\text{def}}{=} & \exists \overline{a^{\text{spec}}_{\text{ct}}}, \overline{a^{\text{spec}}_{\text{mem}}} \approx^{specmem}_{specct} \overline{a^{\text{spec}}_{\text{ct}}} \text{ and } \overline{a^{\text{spec}}_{\text{ct}}} \approx^{specct}_{specarch} \overline{a^{\text{spec}}_{\text{arch}}}
\end{array}$$

3.3 Observer Hierarchy



4 Machines

$$\text{(Fencing)} \quad - \quad \begin{array}{rcl} \mathbf{a}^{\text{spec}}_{\text{fence}} & = & \ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{fence}} \mid \ulcorner \mathbf{Load}\ \ell\ \mathbf{v} \urcorner^{\text{spec}}_{\text{fence}} \mid \ulcorner \mathbf{Store}\ \ell\ \mathbf{v} \urcorner^{\text{spec}}_{\text{fence}} \mid \ulcorner \mathbf{Pc}\ \mathbf{n} \urcorner^{\text{spec}}_{\text{fence}} \\ & \mid & \ulcorner \mathbf{Spec} \urcorner^{\text{spec}}_{\text{fence}} \mid \ulcorner \mathbf{Rlb} \urcorner^{\text{spec}}_{\text{fence}} \mid \ulcorner \mathbf{Fence} \urcorner^{\text{spec}}_{\text{fence}} \end{array}$$

$$\text{(Speculative Load Hardening)} \quad - \quad \begin{array}{rcl} \mathbf{a}^{\text{spec}}_{\text{slh}} & = & \ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{slh}} \mid \ulcorner \mathbf{Load}\ \ell\ \mathbf{v} \urcorner^{\text{spec}}_{\text{slh}} \mid \ulcorner \mathbf{Store}\ \ell\ \mathbf{v} \urcorner^{\text{spec}}_{\text{slh}} \mid \ulcorner \mathbf{Pc}\ \mathbf{n} \urcorner^{\text{spec}}_{\text{slh}} \mid \\ & \mid & \ulcorner \mathbf{Br}\ \mathbf{b}\ \mathbf{n} \urcorner^{\text{spec}}_{\text{slh}} \mid \ulcorner \mathbf{Spec} \urcorner^{\text{spec}}_{\text{slh}} \mid \ulcorner \mathbf{Rlb} \urcorner^{\text{spec}}_{\text{slh}} \end{array}$$

5 Mappings from Observers to Machines

Whenever the following trace-relations are applied at top-level, we assume that both sides are filtered with regards to ε.

Memory fencing

Let $o \in \mathbb{O}$ be a stream of „yes" and „no", which associates to each $a^{\text{spec}}_{\text{arch}}$ in a given trace $\overline{a^{\text{spec}}_{\text{arch}}}$ The answer $o(n)$ corresponds to the event at position n in $\overline{a^{\text{spec}}_{\text{arch}}}$.

— $\boxed{\delta^{\text{specarch}}_{\text{specfence}} : \ulcorner \ell \urcorner^{\text{spec}}_{\text{arch}} \to \ulcorner \ell \urcorner^{\text{spec}}_{\text{fence}}}$ „Total function between locations across speculative architecture and speculative fence events." —

$\boxed{\sim^{specarch\,(n)}_{specfence\,(o)} : \mathbb{N} \to \mathbb{O} \to a^{\text{spec}}_{\text{arch}} \to \bar{\mathbf{a}}^{\text{spec}}_{\text{fence}} \to \mathbb{P}}$ „Map speculative architecture to memory-fencing using oracle o for a given position n."

$$\frac{}{\ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{arch}} \sim^{specarch\,(n)}_{specfence\,(o)} \ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{fence}}} \quad \text{(specarch-to-fencing-empty)}$$

$$\frac{}{\ulcorner \lightning \urcorner^{\text{spec}}_{\text{arch}} \sim^{specarch\,(n)}_{specfence\,(o)} \ulcorner \lightning \urcorner^{\text{spec}}_{\text{fence}}} \quad \text{(specarch-to-fencing-crash)}$$

$$\frac{o(n) = \text{yes} \qquad \delta^{\text{specarch}}_{\text{specfence}}(\ulcorner \ell \urcorner^{\text{spec}}_{\text{arch}}) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{fence}}}{\ulcorner Load\ \ell\ v \urcorner^{\text{spec}}_{\text{arch}} \sim^{specarch\,(n)}_{specfence\,(o)} \ulcorner \mathbf{Fence} \urcorner^{\text{spec}}_{\text{fence}} \cdot \ulcorner \mathbf{Load}\ \ell\ \mathbf{v} \urcorner^{\text{spec}}_{\text{fence}}} \quad \text{(specarch-to-fencing-load-yes)}$$

$$\frac{o\,(n) = \text{yes} \qquad \delta^{\text{specarch}}_{\text{specfence}}\left(\ulcorner \ell \urcorner^{\text{spec}}_{\text{arch}}\right) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{fence}}}{\ulcorner \mathit{Store}\ \ell\ v \urcorner^{\text{spec}}_{\text{arch}} \sim^{\mathit{specarch}\,(n)}_{\mathit{specfence}\,(o)} \ulcorner \mathbf{Fence} \urcorner^{\text{spec}}_{\text{fence}} \cdot \ulcorner \mathbf{Store}\ \ell\ \mathbf{v} \urcorner^{\text{spec}}_{\text{fence}}} \quad \text{(specarch-to-fencing-store-yes)}$$

$$\frac{o\,(n) = \text{no} \qquad \delta^{\text{specarch}}_{\text{specfence}}\left(\ulcorner \ell \urcorner^{\text{spec}}_{\text{arch}}\right) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{fence}}}{\ulcorner \mathit{Load}\ \ell\ v \urcorner^{\text{spec}}_{\text{arch}} \sim^{\mathit{specarch}\,(n)}_{\mathit{specfence}\,(o)} \ulcorner \mathbf{Load}\ \ell\ \mathbf{v} \urcorner^{\text{spec}}_{\text{fence}}} \quad \text{(specarch-to-fencing-load-no)}$$

$$\frac{o\,(n) = \text{no} \qquad \delta^{\text{specarch}}_{\text{specfence}}\left(\ulcorner \ell \urcorner^{\text{spec}}_{\text{arch}}\right) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{fence}}}{\ulcorner \mathit{Store}\ \ell\ v \urcorner^{\text{spec}}_{\text{arch}} \sim^{\mathit{specarch}\,(n)}_{\mathit{specfence}\,(o)} \ulcorner \mathbf{Store}\ \ell\ \mathbf{v} \urcorner^{\text{spec}}_{\text{fence}}} \quad \text{(specarch-to-fencing-store-no)}$$

$$\frac{}{\ulcorner \mathit{Pc}\ n \urcorner^{\text{spec}}_{\text{arch}} \sim^{\mathit{specarch}\,(n)}_{\mathit{specfence}\,(o)} \ulcorner \mathbf{Pc}\ \mathbf{n} \urcorner^{\text{spec}}_{\text{fence}}} \quad \text{(specarch-to-fencing-pc)}$$

$$\frac{}{\ulcorner \mathit{Spec} \urcorner^{\text{spec}}_{\text{arch}} \sim^{\mathit{specarch}\,(n)}_{\mathit{specfence}\,(o)} \ulcorner \mathbf{Spec} \urcorner^{\text{spec}}_{\text{fence}}} \quad \text{(specarch-to-fencing-spec)}$$

$$\frac{}{\ulcorner \mathit{Rlb} \urcorner^{\text{spec}}_{\text{arch}} \sim^{\mathit{specarch}\,(n)}_{\mathit{specfence}\,(o)} \ulcorner \mathbf{Rlb} \urcorner^{\text{spec}}_{\text{fence}}} \quad \text{(specarch-to-fencing-rlb)}$$

$\boxed{\sim^{\mathit{specarch}\,(n)}_{\mathit{specfence}\,(o)} : \mathbb{N} \to \mathbb{O} \to a^{\text{spec}}_{\text{arch}} \to \mathbf{a}^{\text{spec}}_{\text{fence}} \to \mathbb{P}}$ „Map speculative architecture to memory-fencing using oracle o for a given position n."

$$\frac{}{[\cdot] \approx^{\mathit{specarch}\,(n)}_{\mathit{specfence}\,(o)} [\cdot]} \quad \text{(specarch-to-fencing-refl)}$$

$$\frac{a^{\text{spec}}_{\text{arch}} \sim^{\mathit{specarch}\,(n)}_{\mathit{specfence}\,(o)} \overline{\mathbf{a}}^{\text{spec}}_{\text{fence}} \qquad \overline{a^{\text{spec}}_{\text{arch}}} \sim^{\mathit{specarch}\,(1+n)}_{\mathit{specfence}\,(o)} \overline{\mathbf{a}}^{\text{spec}\prime}_{\text{fence}}}{a^{\text{spec}}_{\text{arch}} \cdot \overline{a^{\text{spec}}_{\text{arch}}} \approx^{\mathit{specarch}\,(n)}_{\mathit{specfence}\,(o)} \overline{\mathbf{a}}^{\text{spec}}_{\text{fence}} \cdot \overline{\mathbf{a}}^{\text{spec}\prime}_{\text{fence}}} \quad \text{(specarch-to-fencing-trans)}$$

Speculative Load Hardening

Let $s \in \mathbb{S}$ be a set of secret memory addresses, i.e., we say ℓ is secret iff $\ell \in s$. We write $\overline{\mathbf{b}}$ for a stack of locations ($\mathcal{B}$). Note that we assume speculative execution to use always-mispredict semantics.

$\boxed{\delta^{\text{specarch}}_{\text{specslh}} : \ulcorner \ell \urcorner^{\text{spec}}_{\text{arch}} \to \ulcorner \ell \urcorner^{\text{spec}}_{\text{slh}}}$ „Total function between locations across speculative architecture and speculative slh events."

$\boxed{\sim^{\mathit{specarch}\,(n;s;\overline{\mathbf{b}})}_{\mathit{specslh}\,(m;\overline{\mathbf{b}})} : \mathbb{N} \to \mathbb{S} \to \mathcal{B} \to \mathbb{N} \to \mathcal{B} \to a^{\text{spec}}_{\text{arch}} \to \overline{\mathbf{a}}^{\text{spec}}_{\text{slh}} \to \mathbb{P}}$ „Map speculative architecture to speculative load hardening. s is a set of secret locations. $\overline{\mathbf{b}}$ is a stack of conditions."

$$\frac{}{\ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{arch}} \sim^{\mathit{specarch}\,(n;s;\overline{\mathbf{b}})}_{\mathit{specslh}\,(n;\overline{\mathbf{b}})} \ulcorner \varepsilon \urcorner^{\text{spec}}_{\text{slh}}} \quad \text{(specarch-to-specslh-empty)}$$

$$\frac{}{\ulcorner \lightning \urcorner^{\text{spec}}_{\text{arch}} \sim^{\mathit{specarch}\,(n;s;\overline{\mathbf{b}})}_{\mathit{specslh}\,(n;\overline{\mathbf{b}})} \ulcorner \lightning \urcorner^{\text{spec}}_{\text{slh}}} \quad \text{(specarch-to-specslh-crash)}$$

$$\frac{}{\ulcorner \mathit{Pc}\ n' \urcorner^{\text{spec}}_{\text{arch}} \sim^{\mathit{specarch}\,(1+n;s;\overline{\mathbf{b}})}_{\mathit{specslh}\,(1+n;\mathbf{b}\cdot\overline{\mathbf{b}})} \ulcorner \mathbf{Br}\ \mathbf{b}\ \mathbf{n}' \urcorner^{\text{spec}}_{\text{slh}}} \quad \text{(specarch-to-specslh-specmode-branch)}$$

$$\frac{\ell \in s \qquad \mathbf{v} = v \text{ bitor } \overline{\mathbf{b}} \qquad \delta^{\text{specarch}}_{\text{specslh}}\left(\ulcorner \ell \urcorner^{\text{spec}}_{\text{arch}}\right) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{slh}}}{\ulcorner \mathit{Store}\ \ell\ v \urcorner^{\text{spec}}_{\text{arch}} \sim^{\mathit{specarch}\,(1+n;s;\overline{\mathbf{b}})}_{\mathit{specslh}\,(1+n;\overline{\mathbf{b}})} \ulcorner \mathbf{Store}\ \ell\ \mathbf{v} \urcorner^{\text{spec}}_{\text{slh}}} \quad \text{(specarch-to-specslh-specmode-store)}$$

$$\frac{\ell \in s \qquad \mathbf{v} = v \text{ bitor } \overline{\mathbf{b}} \qquad \delta^{\text{specarch}}_{\text{specslh}}\left(\ulcorner \ell \urcorner^{\text{spec}}_{\text{arch}}\right) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{slh}}}{\ulcorner \mathit{Load}\ \ell\ v \urcorner^{\text{spec}}_{\text{arch}} \sim^{\mathit{specarch}\,(1+n;s;\overline{\mathbf{b}})}_{\mathit{specslh}\,(1+n;\overline{\mathbf{b}})} \ulcorner \mathbf{Load}\ \ell\ \mathbf{v} \urcorner^{\text{spec}}_{\text{slh}}} \quad \text{(specarch-to-specslh-specmode-load)}$$

$$\frac{\delta^{\text{specarch}}_{\text{specslh}}\left(\ulcorner \ell \urcorner^{\text{spec}}_{\text{arch}}\right) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{slh}}}{\ulcorner \mathit{Store}\ \ell\ v \urcorner^{\text{spec}}_{\text{arch}} \sim^{\mathit{specarch}\,(0;s;\overline{\mathbf{b}})}_{\mathit{specslh}\,(0;\overline{\mathbf{b}})} \ulcorner \mathbf{Store}\ \ell\ \mathbf{v} \urcorner^{\text{spec}}_{\text{slh}}} \quad \text{(specarch-to-specslh-store)}$$

$$\frac{\delta^{\text{specarch}}_{\text{specslh}}\left(\ulcorner \ell \urcorner^{\text{spec}}_{\text{arch}}\right) = \ulcorner \ell \urcorner^{\text{spec}}_{\text{slh}}}{\ulcorner \mathit{Load}\ \ell\ v \urcorner^{\text{spec}}_{\text{arch}} \sim^{\mathit{specarch}\ (0;s;\overline{\mathbf{b}})}_{\mathit{specslh}\ (0;\overline{\mathbf{b}})} \ulcorner \mathbf{Load}\ \ell\ \mathbf{v} \urcorner^{\text{spec}}_{\text{slh}}} \quad \text{(specarch-to-specslh-load)}$$

$$\frac{}{\ulcorner \mathit{Spec} \urcorner^{\text{spec}}_{\text{arch}} \sim^{\mathit{specarch}\ (n;s;\overline{\mathbf{b}})}_{\mathit{specslh}\ (1+n;\overline{\mathbf{b}})} \ulcorner \mathbf{Spec} \urcorner^{\text{spec}}_{\text{slh}}} \quad \text{(specarch-to-specslh-spec)}$$

$$\frac{}{\ulcorner \mathit{Rlb} \urcorner^{\text{spec}}_{\text{arch}} \sim^{\mathit{specarch}\ (1+n;s;[\cdot])}_{\mathit{specslh}\ (n;[\cdot])} \ulcorner \mathbf{Rlb} \urcorner^{\text{spec}}_{\text{slh}}} \quad \text{(specarch-to-specslh-rlb-nop)}$$

$$\frac{}{\ulcorner \mathit{Rlb} \urcorner^{\text{spec}}_{\text{arch}} \sim^{\mathit{specarch}\ (1+n;s;\mathbf{b}\cdot\overline{\mathbf{b}})}_{\mathit{specslh}\ (n;\overline{\mathbf{b}})} \ulcorner \mathbf{Rlb} \urcorner^{\text{spec}}_{\text{slh}}} \quad \text{(specarch-to-specslh-rlb-pop)}$$

$$\frac{}{\ulcorner \mathit{Pc}\ n' \urcorner^{\text{spec}}_{\text{arch}} \sim^{\mathit{specarch}\ (n;s;\overline{\mathbf{b}})}_{\mathit{specslh}\ (n;\overline{\mathbf{b}})} \ulcorner \mathbf{Pc}\ \mathbf{n}' \urcorner^{\text{spec}}_{\text{slh}}} \quad \text{(specarch-to-specslh-pc)}$$

$$\frac{}{\ulcorner \mathit{Pc}\ n' \urcorner^{\text{spec}}_{\text{arch}} \sim^{\mathit{specarch}\ (0;s;\overline{\mathbf{b}})}_{\mathit{specslh}\ (n;\overline{\mathbf{b}})} \ulcorner \mathbf{Branch}\ \mathbf{b}'\ \mathbf{n}' \urcorner^{\text{spec}}_{\text{slh}}} \quad \text{(specarch-to-specslh-branch)}$$

$$\boxed{\sim^{\mathit{specarch}\ (n;s;\overline{\mathbf{b}})}_{\mathit{specslh}\ (m)} : \mathbb{N} \to \mathbb{S} \to \mathcal{B} \to \mathbb{N} \to a^{\text{spec}}_{\text{arch}} \to \overline{\mathbf{a}}^{\text{spec}}_{\text{slh}} \to \mathbb{P}}$$

„Map speculative architecture to speculative load hardening. s is a set of secret locations. $\overline{\mathbf{b}}$ is a stack of conditions."

$$\frac{}{[\cdot] \approx^{\mathit{specarch}\ (0;s;[\cdot])}_{\mathit{specslh}\ (0)} [\cdot]} \quad \text{(specarch-to-specslh-refl)}$$

$$\frac{a^{\text{spec}}_{\text{arch}} \sim^{\mathit{specarch}\ (n;s;\mathbf{b})}_{\mathit{specslh}\ (n';\mathbf{b}')} \mathbf{a}^{\text{spec}}_{\text{slh}} \qquad \overline{a^{\text{spec}}_{\text{arch}}} \approx^{\mathit{specarch}\ (n';s;\mathbf{b}')}_{\mathit{specslh}\ (m)} \overline{\mathbf{a}}^{\text{spec}}_{\text{slh}}}{a^{\text{spec}}_{\text{arch}} \cdot \overline{a^{\text{spec}}_{\text{arch}}} \approx^{\mathit{specarch}\ (n;s;\mathbf{b})}_{\mathit{specslh}\ (m)} \mathbf{a}^{\text{spec}}_{\text{slh}} \cdot \overline{\mathbf{a}}^{\text{spec}}_{\text{slh}}} \quad \text{(specarch-to-specslh-trans)}$$

6 Combinations

For a trace $\overline{a}$, let $\overline{a_\circ}$ be the *input* and $\overline{a_\bullet}$ be the *output* projections. The input and output projections are assumed to be disjoint. Let ϕ be a selection abstraction in the style of abstract non-interference. Let $[\![\bullet]\!]$ be an observer and $\{\overline{a}\}_{\text{low}}$ be the equivalence class for a trace $\overline{a}$ using the standard low-equality notion, equating only public events and ignoring differences in private events.

▷ **DEFINITION** 1 (Observer-specific non-interference).

$$\text{oNI} \stackrel{\text{def}}{=} \{\pi \mid \forall \overline{a}, \overline{a'} \in \pi, \text{if } \phi\left(\overline{a_\circ}\right) = \phi\left(\overline{a_\circ}'\right), \text{ then } [\![\overline{a_\bullet}]\!] = [\![\overline{a_\bullet}']\!]\}$$

Definition 1 (Observer-specific non-interference) can be instantiated for all previously discussed observer models (Sections 2 and 4).

Combining Fencing and SLH

Glossary

$[\![\cdot]\!]^{\textbf{spec}}_{\textbf{arch}}$ SPEC-execution; ARCHobs-observations. 1

$[\![\cdot]\!]^{\mathbf{spec}}_{\mathbf{ct}}$ SPEC-execution; CTobs-observations.. 1

$[\![\cdot]\!]^{\mathbf{spec}}_{\mathbf{mem}}$ SPEC-execution; MEMobs-observations.. 1

ADD Artificial Data Dependencies. 1

ARCHobs Architectural observer. Sees any microarchitectural covert channel.. 1, 11

CFI Control-Flow Integrity. 1

CMR Cancelable Memory Requests. 1

CTobs Sees loads, stores, and program counter memory addresses.. 1, 12

DTT Dynamic Taint Tracking. 1

FI Fence Insertion. 1

H Hardware. 1

HS Hardware-Simulation. 1

ISO Process Isolation. 1

MEMobs Sees loads and store memory addresses.. 1, 12

MSR Model-Specific Register. 1

NTMMap Non-Transient Memory Mappings. 1

PATH-CTobs Constant-Time observer. Sees loads, stores, and program counter memory addresses, as well as all possible (speculative) paths.. 1

RI Register Interlocking. 1

RZ Register Zeroing. 1

S Software. 1

SEQ Sequential, in-order execution of instructions.. 1

SEQOOO Sequential, out-of-order execution of instructions.. 1

SLH Speculative Load Hardening. 1

SM Shadow Memory. 1

SPEC Speculative, in-order execution of instructions.. 1, 11, 12

SPECOOO Speculative, out-of-order execution of instructions.. 1

SSLH Strong Speculative Load Hardening. 1

USLH Ultimate Speculative Load Hardening. 1

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